

Research Evidence, Data, Sources & References Document

4.1 Research Objective

This research investigated the following dimensions to establish the problem, validate the solution, and assess feasibility for Project Saarthi:

1. **Tourist problems** — What do visitors actually face on-site at Indian heritage monuments?
 2. **Existing solutions** — What platforms, apps, and systems currently exist, and where do they fall short?
 3. **Technology** — What technologies are available and mature enough to power Saarthi?
 4. **Feasibility** — Is Saarthi technically, economically, and operationally viable?
 5. **User needs** — What information do tourists need, in what format, and at what moment?
 6. **Heritage management** — How do ASI, UNESCO, and state bodies manage and disseminate heritage information?
 7. **Economic impact** — What is the scale of heritage tourism and the potential for local economic inclusion?
 8. **Local communities** — How are artisans, guides, and residents currently excluded from digital tourism?
 9. **Accessibility** — What barriers do visitors with disabilities face, and how can digital tools help?
 10. **Digital heritage** — What global and national best practices exist for digital heritage interpretation?
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4.2 Problem Research

ID	Finding	Data / Evidence	Source	Relevance to Saarthi
P01	Official heritage information is insufficient	<50% of surveyed visitors considered ASI website information adequate; on-site ASI info rated "good quality but insufficient quantity"	Survey evidence — <i>source verification required</i>	Saarthi requires a more comprehensive verified information layer beyond what ASI currently provides digitally
P02	Heavy dependence on online sources with misinformation risk	Internet is main info source for ~50.1% domestic and ~73.8% foreign tourists; search results often surface commercial sites before official pages	Ministry of Tourism study — <i>source verification required</i>	Saarthi must provide a verified, trustworthy alternative that outranks commercial misinformation
P03	Lesser-known sites are poorly covered	Foreign visitors knew about only ~two-thirds of less-visited monuments; these sites have weaker signage and digital interpretation	Survey evidence — <i>source verification required</i>	Saarthi's AI discovery and recommendation engine must actively surface lesser-known sites
P04	Contextual interpretation is weak	Many sites provide basic names/dates but not historical phases, architecture, cultural meaning, or community context	Government reports — <i>source verification required</i>	Saarthi's verified knowledge layer must go beyond facts to deliver deep contextual storytelling
P05	Navigation and signage gaps	Directional signs and information boards often missing, unclear, or badly placed, especially at less-visited monuments	Government reports — <i>source verification required</i>	Saarthi's GPS + geofencing + QR navigation must fill physical signage gaps
P06	Language and accessibility barriers	Multilingual signs, audio guides, and accessible-route information inconsistent; visitors with disabilities face missing infrastructure	Survey evidence — <i>source verification required</i>	Saarthi's multilingual voice-first design and accessible routing are core differentiators

P07	Misinformation and unreliable guides	Tourists rely on unverified websites, social media, touts, and informal guides when official sources are weak	Survey evidence — <i>source verification required</i>	Saarthi's verified content pipeline and guide authentication address this directly
P08	Authenticity not clarified	Visitors not told whether features are original, restored, reconstructed, or interpretive	ICOMOS guidance	Saarthi's authenticity labels (original/restored/reconstructed) fill a documented gap
P09	Travel scams are widespread	~47% of Indian travelers report travel scams	Survey evidence — <i>source verification required</i>	Saarthi's verified information and guide authentication reduce scam exposure
P10	Wrong heritage information is common	~53% of travelers report receiving incorrect heritage information	Survey evidence — <i>source verification required</i>	Saarthi's source-grounded RAG pipeline with citations directly counters misinformation
P11	Meaningful experience gap	~74% seek meaningful experiences; only ~10% receive them; 40–60% failure rate at gateways	Global/experiential benchmark — <i>source verification required</i>	Saarthi's contextual, personalized storytelling is designed to close this gap
P12	Visitor barriers	31.8% domestic / 23.3% foreign cite entry fees; 29.9% / 21.1% cite time; 13.8% / 12.9% cite itinerary exclusion	Survey evidence — <i>source verification required</i>	Saarthi's AI planner addresses time and itinerary barriers; marketplace addresses cost through local options

4.3 Existing Solutions Research

Platform / System	What Exists	What It Solves	What It Does Not Solve	Saarthi Opportunity
Incredible India Digital Platform	Central travel-information system by Ministry of Tourism	Destination discovery, general travel planning	On-site interpretation, location-triggered content, personalization, offline access	Saarthi provides the on-site, location-aware layer that Incredible India does not

ASI Website & e-Ticketing	Monument lists, World Heritage info, visitor instructions, e-ticketing for select monuments	Official information access, digital ticketing	Depth of content (<50% adequacy), multilingual support, on-site navigation, personalization	Saarthi ingests ASI data and enriches it with contextual, multilingual, location-aware delivery
Indian Heritage App (ASI / Adopt a Heritage 2.0)	Geo-tagged monument locations, photographs, public amenities, maps, basic information	Basic site discovery and amenity information	Deep contextual interpretation, multilingual audio, AI personalization, offline capability, AR	Saarthi extends beyond amenity listing to immersive, intelligent interpretation
Audio Odigos (Ministry of Tourism)	Audio explanations, site maps, images, historical content for 12 sites; downloadable chapters	Self-guided audio exploration at select sites	Limited to 12 sites; no real-time triggering; no personalization; no navigation; no AR	Saarthi scales audio interpretation to all monuments with AI personalization and location triggering
QR-based Systems	QR codes at monuments linking to text, photos, videos, audio	Physical-to-digital connection at specific points	Internet-dependent; static content; no personalization; no navigation beyond the QR point	Saarthi integrates QR as one of multiple triggering mechanisms within a unified offline-capable platform
Google Maps / Street View	Locations, routes, photographs, 360° imagery	Navigation and pre-visit visual planning	No heritage interpretation; no multilingual audio; no accessibility routing; no real-time info	Saarthi uses maps APIs for navigation but adds the entire heritage interpretation layer
Google Arts & Culture	Online heritage stories, high-res images, virtual tours, thematic projects (Ajanta–Ellora, Wonders of India)	Digital cultural storytelling and virtual access	Pre-visit/remote focused; no on-site location awareness; no personalization; no offline mode	Saarthi brings similar richness to the physical on-site experience

Kerala Tourism AR Heritage Walk	AR overlays around Sree Padmanabhaswamy Temple and heritage areas	Immersive AR interpretation at specific sites	Limited to select locations; requires strong connectivity and compatible devices; no offline mode	Saarthi integrates AR as an optional feature within a broader offline-capable platform
Government VR/3D Projects	3D laser scanning, virtual tours, holograms, digital installations	Reconstruction of damaged/missing structures; virtual access	Pre-visit focused; not location-aware on-site; not personalized; high infrastructure requirements	Saarthi uses 3D assets for on-site AR with graceful degradation
Indian Culture Portal	Manuscripts, museum collections, photographs, archives, festivals, crafts, music, food traditions	Digital research and cultural heritage access	Research-oriented; not designed for on-site tourist use; no navigation or personalization	Saarthi can draw from this repository to enrich cultural context
State Tourism Platforms (Delhi, Maharashtra, etc.)	Heritage routes, monument information, audio/video guides, VR, AI initiatives	Local destination information and some digital interpretation	Fragmented across states; inconsistent quality; no unified national platform; limited offline	Saarthi provides a unified cross-state platform with consistent quality
Booking-first OTAs (MakeMyTrip, EaseMyTrip, IRCTC)	Hotel, ticket, and package bookings	Transaction efficiency	No heritage education; no on-site interpretation; no personalization beyond booking	Saarthi fills the entire interpretation and exploration gap that OTAs ignore
Informal Guide Ecosystem	On-site narration by licensed and unlicensed guides	Human-led interpretation and navigation	Static scripts; uneven training; unverifiable credentials; misinformation risk; no multilingual consistency	Saarthi's guide verification and AI-augmented content support and improve the guide ecosystem

4.4 Technology Research

Technology	Research Finding	Maturity	Saarthi Application
AI / LLM	LLMs can generate natural-language explanations, summarize complex histories, and support multilingual interaction	Production-ready	Heritage explanations, chat interaction, content generation
Agentic AI	Multi-agent workflows can handle complex planning tasks involving multiple constraints	Emerging but viable	AI Journey Planner with budget, time, interest, weather, crowd constraints
RAG	Retrieval-Augmented Generation grounds AI outputs in verified source documents, reducing hallucination	Production-ready	Source-grounded heritage explanations with citations
Recommendation Systems	Collaborative and content-based filtering can personalize content and route suggestions	Mature	Personalized site recommendations and itinerary generation
GPS	Smartphone GPS provides 3–10 m accuracy under open sky; dual-frequency (L1+L5) reaches 1–3 m	Ubiquitous	Zone-level content triggering and navigation
AR (ARCore/ARKit)	1,200+ Android models support ARCore 1.40+ by 2026; tracking works well in open heritage sites but degrades in low light and cluttered interiors	Mature for flagship devices	Historical reconstruction overlays, structure recognition
3D / Photogrammetry	3D scanning and photogrammetry can create accurate digital twins of heritage structures	Established	AR assets, virtual reconstruction, architectural visualization
Computer Vision	Object and structure recognition can identify specific sculptures, inscriptions, and architectural elements	Production-ready	AR content triggering via camera
Multilingual AI	Neural machine translation achieves high quality for major languages; Indic-focused APIs (Sarvam, Bhashini) improving rapidly	Good for major languages; improving for others	Content translation, real-time chat translation
TTS / STT	Text-to-speech and speech-to-text are mature for major languages; Indic	Mature	Voice-first heritage narration and speech

Technology	Research Finding	Maturity	Saarthi Application
	language support expanding		queries
Maps APIs	Google Maps and Mapbox provide comprehensive routing, offline tiles, and navigation	Mature	On-site navigation, facility routing, last-mile guidance
Weather APIs	OpenWeatherMap, AccuWeather, Google Weather provide reliable forecasts	Mature	Weather-aware itinerary suggestions
Cloud Infrastructure	AWS, Azure, GCP all have India regions; managed services reduce operational overhead	Mature	Backend hosting, CDN, storage, AI inference
Offline Technology	Offline-first architectures with downloadable content packs and sync-on-reconnect are well-established	Mature	Site packs for low-connectivity monuments

4.5 Feasibility Research

A. Technical Feasibility

Aspect	Finding	Verdict
Smartphone & ARCore	1,200+ Android models support ARCore by 2026; India's 152M annual shipments dominated by mid-tier Android; AR needs ≥ 4 GB RAM	Feasible with graceful degradation for low-end devices
GPS Accuracy	3–10 m under open sky; 1–3 m with dual-frequency; sufficient for zone-level triggers; hybrid positioning (GPS + QR + map) needed for structure-level	Feasible with hybrid approach
Internet vs. Offline	4G available at popular sites but uneven at remote/rural monuments; offline-first architecture with downloadable site packs is proven	Feasible and essential
AR Requirements	ARCore visual-inertial odometry works in open monuments; battery and data concerns require optimization and low-power alternatives	Feasible as optional feature for flagship sites

Backend Infrastructure	Cloud options in India (AWS Mumbai, Azure Pune, GCP Mumbai); MVP at ₹25K–80K/month; production at ₹40K–1.5L+/month	Feasible with cost optimization
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B. Economic Feasibility

Aspect	Finding	Verdict
Development Cost (MVP)	\$50K–\$150K (₹4–12 mn) for recommendations + multilingual chat + basic maps/audio + limited AR	Feasible with government/CSR funding
Development Cost (Full)	\$250K–\$600K+ (₹20–50 mn+) for multi-agent planner, 3D twins, CV, extensive AR, 50+ monuments	Feasible with phased investment
API & Hosting	Cloud ₹40K–1.2L/month; Maps ₹50K–2L+/month at scale; Translation \$10–20/million chars; Weather ₹3K–40K/month	Feasible with caching and offline modes
Content Creation	3D scanning £3K–10K+ per project; audio ₹0.5–1/word; narrative content requires historians	Major cost center — prioritize high-footfall sites first
Maintenance	15–20% of dev cost/year for software; periodic content reviews; scaling cloud costs	Sustainable with government/PPP funding

C. Operational Feasibility

Aspect	Finding	Verdict
Data Custodianship	ASI (~3,685 sites), state departments, Monument Mitras are established custodians	Feasible — Saarthi aggregates, does not own
Verification Model	Expert-verified core facts + labeled community narratives + RAG-grounded AI	Feasible — mirrors existing Adopt a Heritage workflows
CMS Workflow	Version-controlled CMS with ASI/state editor access, metadata, and approval stages	Feasible — standard content management patterns
Monument Onboarding	Nomination → data collection → prioritization → content production → review → launch	Feasible — mirrors Audio Odigos and Adopt a Heritage 2.0 rollouts

D. Scalability Feasibility

- Phased rollout: Start with 5–10 high-footfall monuments → expand to 50+ → national coverage.
- Content creation is the primary scaling bottleneck; prioritize by footfall and significance.

- Cloud infrastructure scales elastically; cost optimization through caching, offline modes, and multi-vendor inference.

E. Device Feasibility

- Basic mode (text + audio + 2D maps) works on 2 GB RAM Android 10+ devices.
- AR mode requires 4 GB+ RAM and ARCore compatibility (1,200+ models by 2026).
- Graceful degradation ensures no user is excluded.

F. Connectivity Feasibility

- Offline-first architecture eliminates dependency on real-time connectivity.
- Site packs downloadable over Wi-Fi before visits.
- Sync-on-reconnect for updates and analytics.

G. Data Feasibility

- ASI, UNESCO, state archaeology, and Incredible India Content Hub provide substantial existing data.
- Indian Culture Portal offers manuscripts, collections, and intangible heritage.
- Gap: Structure-level narratives, multilingual audio, and 3D assets require new content creation.

4.6 Impact & Benefits Research

Impact Area	Expected Benefit	Supporting Evidence
Visitor Experience	Deeper, more meaningful on-site engagement	74% seek meaningful experiences; only ~10% receive them — Saarthi targets this gap
Learning & Education	Improved historical, architectural, and cultural understanding through contextual storytelling	Government reports note widespread lack of basic interpretation at monuments
Cultural Awareness	Greater appreciation of multiple perspectives, living traditions, and community narratives	UNESCO framework connects visitor experience with cultural understanding
Accessibility	Inclusive access for visitors with disabilities, language barriers, and limited mobility	Current platforms lack consistent accessibility features

Lesser-Known Site Discovery	Increased footfall and economic activity at under-visited monuments	Foreign visitors knew only ~two-thirds of less-visited sites; Saarthi's AI recommendations address this
Local Employment	New income opportunities for verified guides, artisans, and cultural performers	Current digital platforms systematically exclude local stakeholders
Artisan Revenue	Direct digital visibility and marketplace access for craftspeople near heritage clusters	Artisans currently excluded from formal tourism value chains
Homestays	Increased occupancy for local homestays through integrated listings	Homestays lack digital visibility on current heritage platforms
Guide Ecosystem	Improved guide quality through verification, credentialing, and AI-augmented knowledge	Current guide system suffers from static scripts and uneven training
Crowd Management	Reduced congestion at popular sites through dynamic routing and time-slot distribution	UNESCO framework connects visitor management with heritage protection
Conservation	Better visitor behavior through conservation alerts and responsible tourism guidance	ICOMOS guidance on distinguishing evidence vs. reconstruction
Sustainable Tourism	Balanced distribution of tourism benefits across sites and communities	UNESCO Sustainable Tourism Toolkit covers governance, community engagement, visitor behavior, and monitoring

4.7 Research Gap

Central Documented Gap:

Existing solutions provide individual capabilities such as discovery, ticketing, mapping, audio interpretation, AR, or virtual viewing. The research indicates a clear opportunity to integrate **verified, contextual, personalized, multilingual, location-specific, accessible, updated, and offline** information into a **unified on-site heritage experience**.

This gap is explicitly highlighted across:

- Government evaluations (Ministry of Tourism, ASI)

- Academic analyses of digital heritage platforms
- Platform reviews of existing apps and systems
- UNESCO's framework connecting visitor experience, heritage protection, and community benefits

No existing platform in India simultaneously delivers all eight of these attributes at the moment of physical site exploration. Saarthi is designed to fill this specific gap.

4.8 References

Note: The following references are compiled from the project's research data. Entries marked with an asterisk (*) require verification of the exact URL and access date before formal publication.

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